PART I: TEMPERATURE, HEAT AND THE FIRST LAW OF THERMODYNAMICS

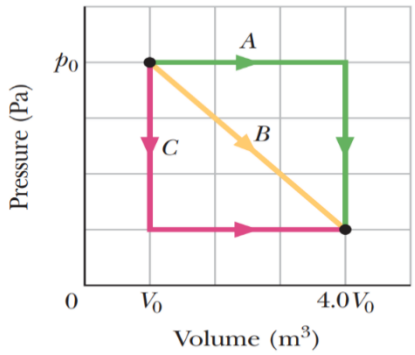
1. At 20°C, a rod is exactly 20.05 cm long on a steel ruler. Both are placed in an oven at 270°C, where the rod now measures 20.11 cm on the same ruler. What is the coefficient of linear expansion for the material of which the rod is made? The coefficient of linear expansion of steel is 11×10^{-6}/°C.

2 A certain substance has a mass per mole of 50.0 g/mol. When 314 J is added as heat to a 30.0 g sample, the sample's temperature rises from 25.0°C to 45.0°C. What are:

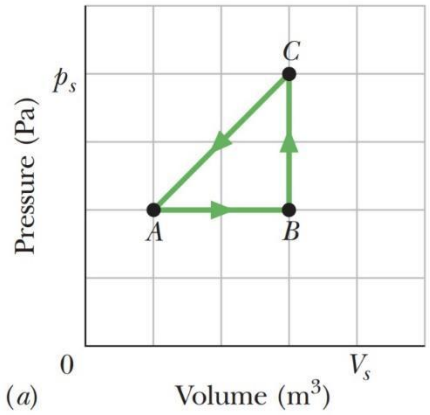
(a) The specific heat of this substance?
(b) The molar specific heat of this substance?
(c) How many moles are in the sample?

3 What mass of steam at 100°C must be mixed with 150 g of ice at its melting point, in a thermally insulated container, to produce liquid water at 50°C?

4 In Fig. below, a gas sample expands from V_0 to $4.0V_0$ while its pressure decreases from p_0 to $p_0/4.0$. If $V_0 = 1.0$ m^3 and $p_0 = 40$ Pa, how much work is done by the gas if its pressure changes with volume via (a) path *A*, (b) path *B*, and (c) path *C*?



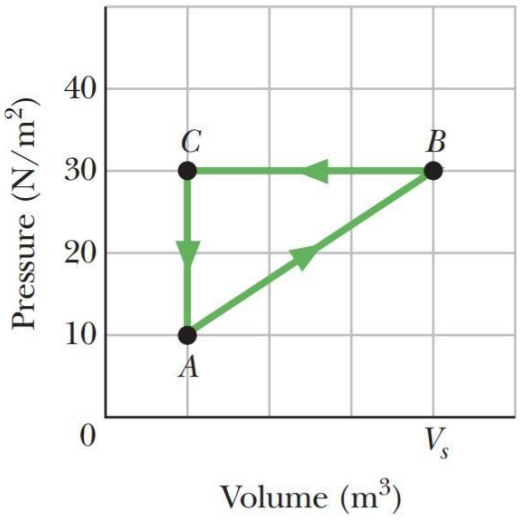
5 A thermodynamic system is taken from state *A* to state *B* to state *C*, and then back to A, as shown in the *p*-*V* diagram of Fig. below (a). The vertical scale is set by $p_s = 40$ Pa, and the horizontal scale is set by $Vs = 4.0$ m^3. (a)–(g) Complete the table in Fig. below (b) by inserting a plus sign, a minus sign, or a zero in each indicated cell. (h) What is the net work done by the system as it moves once through the cycle *ABCA*?



(a) (b)

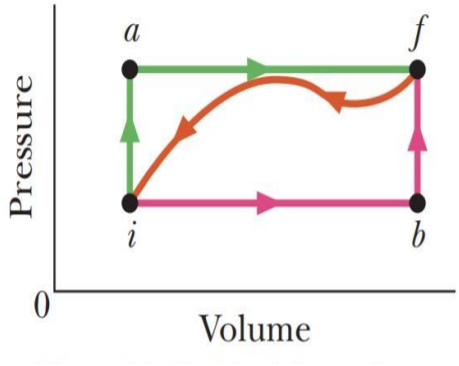
	Q	W	ΔE_{int}
A → B	(a)	(b)	+
B → C	+	(c)	(d)
C → A	(e)	(f)	(g)

6 A gas within a closed chamber undergoes the cycle shown in the *p*-*V* diagram of Fig. below. The horizontal scale is set by Vs = 4.0 m^3. Calculate the net energy added to the system as heat during one complete cycle.

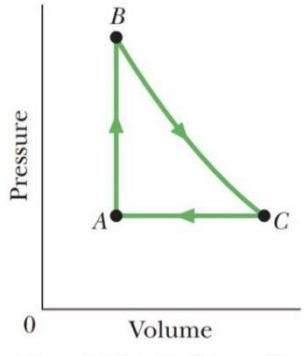


7 Suppose 200 J of work is done on a system and 70.0 cal is extracted from the system as heat. In the sense of the first law of thermodynamics, what are the values (including algebraic signs) of (a) *W*, (b) *Q*, and (c) ΔE_{int}?

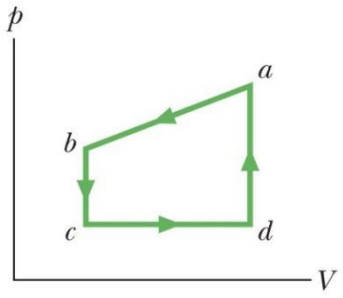
8 When a system is taken from state *i* to state *f* along path *iaf* in **Fig. below**, *Q* = 50 cal and *W* = 20 cal. Along path *ibf*, *Q* = 36 cal. (a) What is *W* along path *ibf*? (b) If *W* = -13 cal for the return path *fi*, what is *Q* for this path? (c) If $E_{int,i}$ = 10 cal, what is $E_{int,f}$? If $E_{int,b}$ = 22 cal, what is *Q* for (d) path *ib* and (e) path *bf* ?



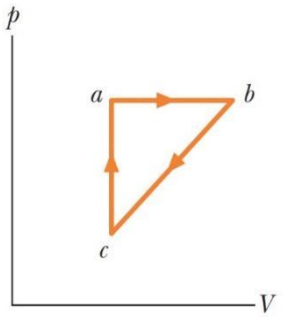
9 As a gas is held within a closed chamber, it passes through the cycle shown in **Fig. below**. Determine the energy transferred by the system as heat during constant-pressure process *CA* if the energy added as heat Q_{AB} during constant-volume process *AB* is 20.0 J, no energy is transferred as heat during adiabatic process *BC*, and the net work done during the cycle is 15.0 J.



10 **Fig. below** represents a closed cycle for a gas (the figure is not drawn to scale). The change in the internal energy of the gas as it moves from *a* to *c* along the path *abc* is -200 J. As it moves from *c* to *d*, 180 J must be transferred to it as heat. An additional transfer of 80 J to it as heat is needed as it moves from *d* to a. How much work is done on the gas as it moves from *c* to *d*?



11 A lab sample of gas is taken through cycle *abca* shown in the p-V diagram of **Fig. below** The net work done is +1.2 J. Along path *ab*, the change in the internal energy is +3.0 J and the magnitude of the work done is 5.0 J. Along path *ca*, the energy transferred to the gas as heat is +2.5 J. How much energy is transferred as heat along (a) path *ab* and (b) path *bc*?



MCQs:

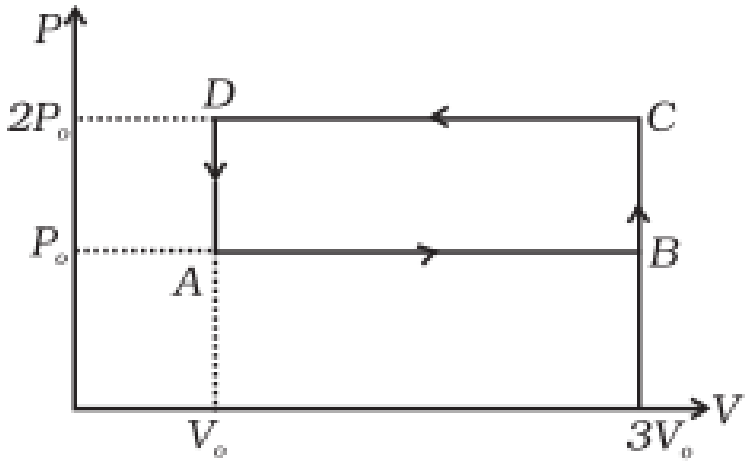
1. Identify intensive (intrinsic) properties:
a. Boiling point, temperature, color
b. Boiling point, temperature, volume
c. Volume, mass, size
d. Length, temperature, mass

2. First law of thermodynamics deals with _____
a. Conservation of mass
b. Conservation of momentum
c. Conservation of energy
d. Conservation of pressure

3. According to First law of thermodynamics,

a. total internal energy of a system during a process remains constant
b. total energy of a system remains constant
c. work done by a system is equal to the heat transferred by the system
d. internal energy, enthalpy and entropy during a process remains constant

4. An ideal gas undergoes the process below. The work done by the gas is



a. - 2PoVo
b. + 2PoVo
c. - 4PoVo
d. + 4 PoVo

5. A system absorbs 50 J of heat and does 11 J of work in a process. The system follows a different path between the same initial and final states and does 15 J work, the heat transferred in the process is:
a. 39 J
b. 65 J
c. 54 J
d. 46 J

6. If 315 cal of heat is given to the system, and the system does 20cal of work, find the change in internal energy
a. 295 cal
b. 335 cal
c. 0 cal
d. 335 J

7. Two cylinders A and B of equal capacity are connected to each other via a stopcock. A contains gas at a standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. If the stopcock is suddenly opened, then the change in internal energy of the gas is:

a. 0
b. 5 J
c. 10 J
d. 15 J

8. Work done on a system is _____, and heat flow into of the system is _____
a. Positive, negative
b. Positive, positive
c. Negative, negative
d. Negative, positive

9. The change internal energy in a cyclic process is _____
a. Zero
b. Infinity
c. Constant
d. Unity

10. The latent heat of vaporisation of water is 2,240 J. If the work done in the process of vaporisation of 1g is 168 J, then the increase in internal energy is?
a. 2408 J

b. 1036 J
c. 4144 J
d. 2072 J